PCA R-Assignment

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2026-10-19

SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(readxl)
library(tidyverse)
library(corrplot)
library(factoextra)
library(psych)
```

2. Import the dataset

```
data <- read_excel("PCA_Customer_Satisfaction data.xlsx", sheet = "Cus
tomer Dataset")
```

Check structure

```
str(data)
```

```
tibble [120 × 9] (S3: tbl_df/tbl/data.frame)
 $ Customer_ID              : chr [1:120] "C001" "C002" "C003" "C004"
...
 $ Product_Quality          : num [1:120] 4.5 4 1.9 3.9 3.3 4 3.3 4.4
3.2 3.5 ...
 $ Staff_Courtesy           : num [1:120] 2.9 3.7 1.7 3.2 2.4 4.4 2.4
2.8 3.6 2.8 ...
 $ Waiting_Time_Satisfaction: num [1:120] 3.1 3 2.3 2.1 3.2 3 2.6 3.2
2.7 3.8 ...
 $ Price_Fairness           : num [1:120] 5 3.8 3.9 4.6 3.8 3 3.9 3.4
2.6 3 ...
 $ Value_for_Money          : num [1:120] 5 4 3.9 4.7 4.8 3.6 4.5 3.3
2.4 3.2 ...
 $ Store_Accessibility      : num [1:120] 1.7 3.8 3.4 2.6 4.1 1 2.5 2.
5 2.9 3.2 ...
 $ Opening_Hours_Suitability: num [1:120] 2.3 3.8 2.5 3.6 3.3 1.4 2.9
3.3 2.3 3.3 ...
 $ Overall_Satisfaction     : num [1:120] 4 4.7 3.2 3.4 3.9 2.5 4 3 3.
3 2.7 ...
```

```
glimpse(data)
```

```
Rows: 120
Columns: 9
$ Customer_ID               <chr> "C001", "C002", "C003", "C004", "C00
5", "C00…
$ Product_Quality           <dbl> 4.5, 4.0, 1.9, 3.9, 3.3, 4.0, 3.3,
4.4, 3.2,…
$ Staff_Courtesy            <dbl> 2.9, 3.7, 1.7, 3.2, 2.4, 4.4, 2.4,
2.8, 3.6,…
$ Waiting_Time_Satisfaction <dbl> 3.1, 3.0, 2.3, 2.1, 3.2, 3.0, 2.6,
3.2, 2.7,…
$ Price_Fairness            <dbl> 5.0, 3.8, 3.9, 4.6, 3.8, 3.0, 3.9,
3.4, 2.6,…
$ Value_for_Money           <dbl> 5.0, 4.0, 3.9, 4.7, 4.8, 3.6, 4.5,
3.3, 2.4,…
$ Store_Accessibility       <dbl> 1.7, 3.8, 3.4, 2.6, 4.1, 1.0, 2.5,
2.5, 2.9,…
$ Opening_Hours_Suitability <dbl> 2.3, 3.8, 2.5, 3.6, 3.3, 1.4, 2.9,
3.3, 2.3,…
$ Overall_Satisfaction      <dbl> 4.0, 4.7, 3.2, 3.4, 3.9, 2.5, 4.0,
3.0, 3.3,…
```

3. Prepare numerical data for PCA

Exclude Customer_ID because it is an identifier, not a rating variable

```
pca_data <- data %>%
    select(-Customer_ID)
```

Check missing values and duplicates

```
colSums(is.na(pca_data))
```

```
          Product_Quality            Staff_Courtesy Waiting_Time_Satis
faction
                        0                         0                  
0
           Price_Fairness           Value_for_Money       Store_Access
ibility
                        0                         0                  
0
Opening_Hours_Suitability      Overall_Satisfaction                  
                        0                         0
```

```
sum(is.na(pca_data))
```

```
[1] 0
```

```
sum(duplicated(pca_data))
```

```
[1] 0
```

```
any(duplicated(data$Customer_ID))
```

```
[1] FALSE
```

4. Descriptive statistics

Alternative: psych::describe

```
describe(pca_data)
```

```
                          vars   n mean   sd median trimmed  mad min m
ax range
Product_Quality              1 120 3.51 0.76    3.6    3.55 0.59 1.8
5.0   3.2
Staff_Courtesy               2 120 3.36 0.76    3.4    3.37 0.74 1.1
4.9   3.8
Waiting_Time_Satisfaction    3 120 3.12 0.80    3.2    3.14 0.89 1.0
4.7   3.7
Price_Fairness               4 120 3.28 0.80    3.3    3.30 0.89 1.0
5.0   4.0
Value_for_Money              5 120 3.44 0.88    3.4    3.45 0.89 1.0
5.0   4.0
Store_Accessibility          6 120 3.11 0.88    3.1    3.10 0.89 1.0
5.0   4.0
Opening_Hours_Suitability    7 120 3.15 0.80    3.2    3.17 0.74 1.0
5.0   4.0
Overall_Satisfaction         8 120 3.26 0.71    3.3    3.25 0.74 1.7
5.0   3.3
                           skew kurtosis   se
Product_Quality           -0.49    -0.37 0.07
Staff_Courtesy            -0.18    -0.39 0.07
Waiting_Time_Satisfaction -0.20    -0.65 0.07
Price_Fairness            -0.26    -0.54 0.07
Value_for_Money           -0.22    -0.32 0.08
Store_Accessibility        0.07    -0.34 0.08
Opening_Hours_Suitability -0.24     0.07 0.07
Overall_Satisfaction       0.07    -0.55 0.06
```

5. Correlation matrix

```
cor_mat <- cor(pca_data, use = "complete.obs")
print(round(cor_mat, 2))
```

```
                          Product_Quality Staff_Courtesy
Product_Quality                      1.00           0.70
Staff_Courtesy                       0.70           1.00
Waiting_Time_Satisfaction            0.47           0.60
Price_Fairness                       0.34           0.10
Value_for_Money                      0.37           0.15
Store_Accessibility                 -0.05           0.05
Opening_Hours_Suitability            0.06           0.21
Overall_Satisfaction                 0.49           0.41
                          Waiting_Time_Satisfaction Price_Fairness
Product_Quality                                0.47           0.34
Staff_Courtesy                                 0.60           0.10
Waiting_Time_Satisfaction                      1.00           0.03
Price_Fairness                                 0.03           1.00
Value_for_Money                                0.01           0.75
Store_Accessibility                            0.42          -0.02
Opening_Hours_Suitability                      0.53          -0.01
Overall_Satisfaction                           0.47           0.52
                          Value_for_Money Store_Accessibility
Product_Quality                      0.37               -0.05
Staff_Courtesy                       0.15                0.05
Waiting_Time_Satisfaction            0.01                0.42
Price_Fairness                       0.75               -0.02
Value_for_Money                      1.00               -0.14
Store_Accessibility                 -0.14                1.00
Opening_Hours_Suitability           -0.05                0.68
Overall_Satisfaction                 0.51                0.28
                          Opening_Hours_Suitability Overall_Satisfacti
on
Product_Quality                            0.06                0.
49
Staff_Courtesy                             0.21                0.
41
Waiting_Time_Satisfaction                  0.53                0.
47
Price_Fairness                            -0.01                0.
52
Value_for_Money                           -0.05                0.
51
Store_Accessibility                        0.68                0.
28
Opening_Hours_Suitability                  1.00                0.
32
Overall_Satisfaction                       0.32                1.
00
```

Visualise correlation matrix

```
corrplot(cor_mat, method = "color", addCoef.col = "black", number.cex
= 0.6, tl.col = "black",
    tl.srt = 45)
```



Identify strongly correlated pairs (e.g., |r| > 0.5)

```
corr_long <- as.data.frame(as.table(cor_mat)) %>%
    filter(Var1 != Var2, abs(Freq) > 0.5) %>%
    arrange(desc(abs(Freq)))
print(corr_long)
```

```
                        Var1                       Var2      Freq
1            Value_for_Money            Price_Fairness 0.7540716
2             Price_Fairness           Value_for_Money 0.7540716
3             Staff_Courtesy           Product_Quality 0.6996000
4            Product_Quality            Staff_Courtesy 0.6996000
5  Opening_Hours_Suitability       Store_Accessibility 0.6839539
6        Store_Accessibility Opening_Hours_Suitability 0.6839539
7  Waiting_Time_Satisfaction            Staff_Courtesy 0.5960774
8             Staff_Courtesy Waiting_Time_Satisfaction 0.5960774
9  Opening_Hours_Suitability Waiting_Time_Satisfaction 0.5334907
10 Waiting_Time_Satisfaction Opening_Hours_Suitability 0.5334907
11      Overall_Satisfaction            Price_Fairness 0.5154520
12            Price_Fairness      Overall_Satisfaction 0.5154520
13      Overall_Satisfaction           Value_for_Money 0.5063173
14           Value_for_Money      Overall_Satisfaction 0.5063173
```

6. Standardise variables

```
pca_scaled <- scale(pca_data)
```

7. Run PCA

```
pca_res <- prcomp(pca_scaled, center = TRUE, scale. = TRUE)
summary(pca_res)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4     PC5     PC6
PC7
Standard deviation     1.7818 1.4392 1.1389 0.61326 0.57858 0.53638 0.
48967
Proportion of Variance 0.3968 0.2589 0.1621 0.04701 0.04184 0.03596 0.
02997
Cumulative Proportion  0.3968 0.6558 0.8179 0.86493 0.90677 0.94273 0.
97271
                           PC8
Standard deviation     0.46727
Proportion of Variance 0.02729
Cumulative Proportion  1.00000
```

8. Eigenvalues and explained variance

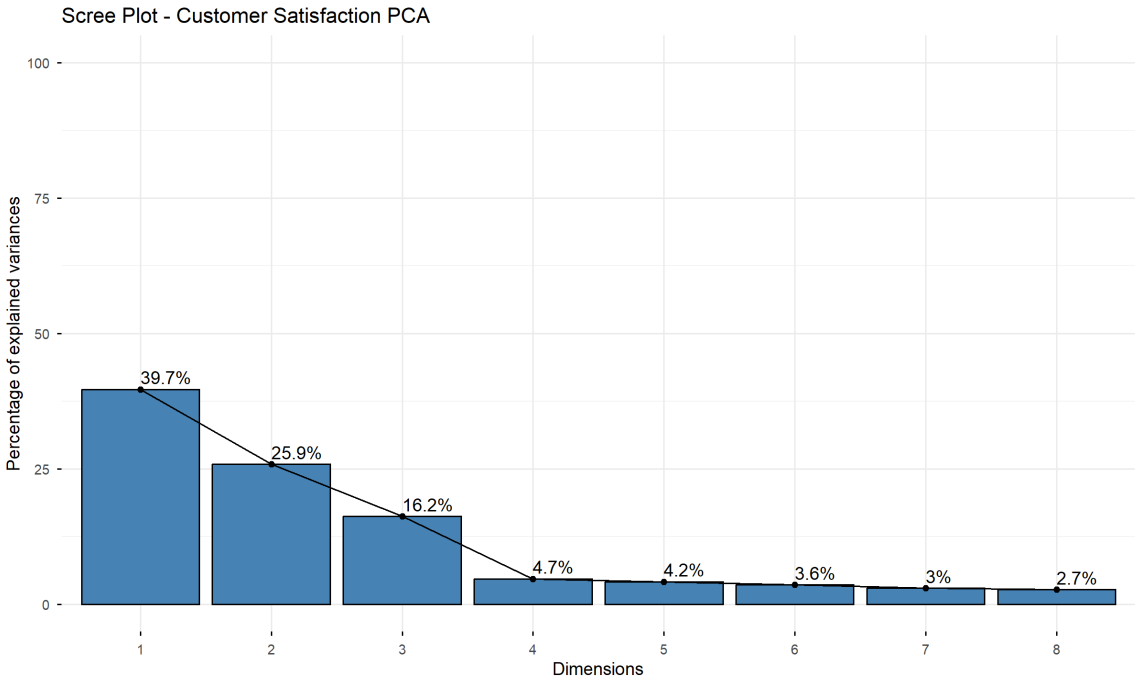
```
eigenvals <- pca_res$sdev^2
explained_var <- eigenvals/sum(eigenvals) * 100
cum_var <- cumsum(explained_var)
eigen_table <- data.frame(Component = paste0("PC", 1:length(eigenval
s)), Eigenvalue = eigenvals,
    Variance_Percent = explained_var, Cumulative_Percent = cum_var)
print(eigen_table)
```

```
  Component Eigenvalue Variance_Percent Cumulative_Percent
1       PC1  3.1747881        39.684852            39.68485
2       PC2  2.0713447        25.891809            65.57666
3       PC3  1.2971883        16.214854            81.79151
4       PC4  0.3760898         4.701123            86.49264
5       PC5  0.3347555         4.184443            90.67708
6       PC6  0.2877085         3.596356            94.27344
7       PC7  0.2397811         2.997263            97.27070
8       PC8  0.2183440         2.729300           100.00000
```

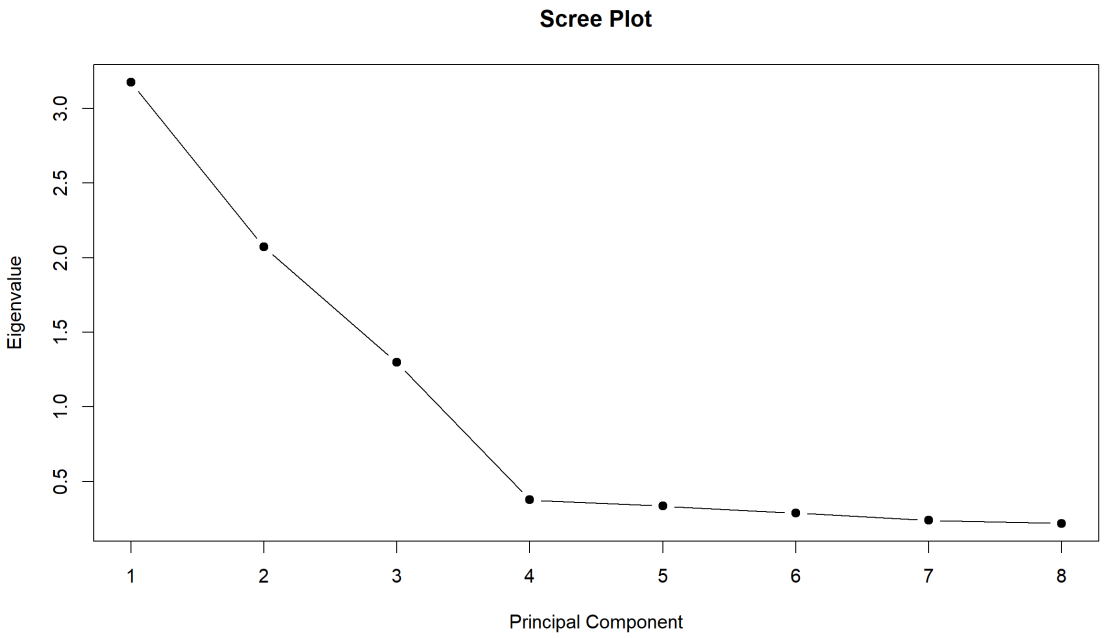
```
kaiser_components <- eigen_table$Component[eigen_table$Eigenvalue > 1]
print(kaiser_components)
```

```
[1] "PC1" "PC2" "PC3"
```

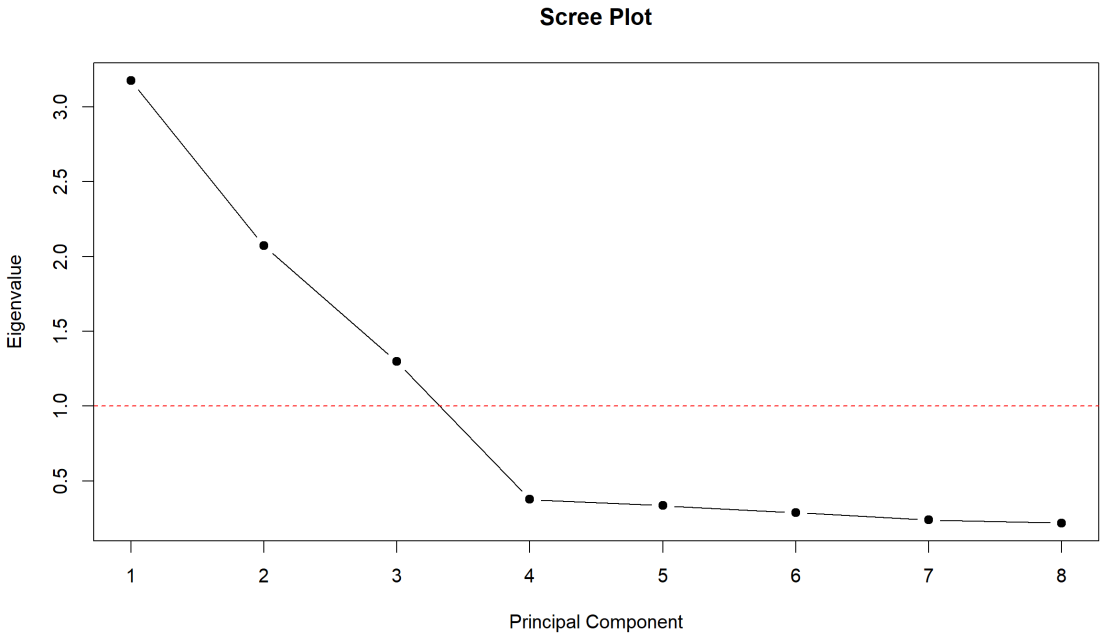
```
fviz_eig(pca_res, addlabels = TRUE, ylim = c(0, 100), barfill = "steel
blue", barcolor = "black") +
    ggtitle("Scree Plot - Customer Satisfaction PCA")
```



```
plot(eigenvals, type = "b", pch = 19, xlab = "Principal Component", yl
ab = "Eigenvalue",
    main = "Scree Plot")
```




```
abline(h = 1, col = "red", lty = 2)
```



10. Save selected components for further analysis

Example: retain first 2 components (adjust based on scree/Kaiser)

```
selected_pcs <- data.frame(Customer_ID = data$Customer_ID, pca_res$x[,
1:3])
write.csv(selected_pcs, "selected_pca_components.csv", row.names = FAL
SE)
```

11. Component loadings

```
loadings <- pca_res$rotation
print(round(loadings[, 1:3], 2))
```

```
                           PC1   PC2   PC3
Product_Quality          -0.42 -0.17  0.41
Staff_Courtesy           -0.40  0.05  0.52
Waiting_Time_Satisfaction -0.41  0.32  0.19
Price_Fairness           -0.30 -0.43 -0.39
Value_for_Money          -0.29 -0.48 -0.29
Store_Accessibility      -0.20  0.48 -0.41
Opening_Hours_Suitability -0.27  0.47 -0.30
Overall_Satisfaction     -0.46 -0.08 -0.19
```

Show largest absolute loadings for first two PCs

```
loadings_df <- as.data.frame(pca_res$rotation) %>%
    rownames_to_column("Variable") %>%
    pivot_longer(-Variable, names_to = "PC", values_to = "Loading") %
>%
    group_by(PC) %>%
    slice_max(abs(Loading), n = 3) %>%
    arrange(PC, desc(abs(Loading)))
print(loadings_df)
```

```
# A tibble: 24 × 3
# Groups:   PC [8]
   Variable                   PC    Loading
   <chr>                      <chr>   <dbl>
 1 Overall_Satisfaction       PC1    -0.461
 2 Product_Quality            PC1    -0.417
 3 Waiting_Time_Satisfaction  PC1    -0.410
 4 Store_Accessibility        PC2     0.483
 5 Value_for_Money            PC2    -0.477
 6 Opening_Hours_Suitability  PC2     0.466
 7 Staff_Courtesy             PC3     0.515
 8 Store_Accessibility        PC3    -0.410
 9 Product_Quality            PC3     0.406
10 Overall_Satisfaction       PC4    -0.747
# i 14 more rows
```

```
fviz_pca_var(pca_res, col.var = "contrib", repel = TRUE) + ggtitle("PC
A Variable Loadings")
```

